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# INFLUENCE OF AGING ON CHONDROITIN SULFATE PROTEOGLYCAN EXPRESSION AND NEURAL STEM/PROGENITOR CELLS IN RAT BRAIN AND IMPROVING EFFECTS OF A HERBAL MEDICINE, *YOKUKANSAN*

**Y. TANAKA AND K. MIZOGUCHI***

*Section of Oriental Medicine, Department of Geriatric Medicine, National Institute for Longevity Sciences, National Center for Geriatrics and Gerontology (NCGG), 36-3 Gengo, Morioka, Obu, Aichi 474–8522, Japan*

**Abstract—There is evidence of structural and functional deterioration in the brain, including the prefrontal cortex (PFC) and hippocampus, during the normal aging process in animals and humans. Extracellular matrix-associated glycoproteins, such as chondroitin sulfate proteoglycans (CSPGs), are involved in not only maintaining the structures and functions of adult neurons, but also regulating the proliferation, migration, and neurite outgrowth of neural stem cells in the brain. On the other hand, a herbal medicine, *yokukansan* (YKS), is used in a variety of clinical situations for treating symptoms associated with age-related neurodegenerative disorders such as Alzheimer's disease, but its pharmacological properties have not been fully understood. The present study was designed to clarify the influence of aging and the improving effects of YKS on the expression of aggrecan, a major molecule of CSPGs, and on the proliferation and migration of neural stem/progenitor cells identified by bromodeoxyuridine (BrdU) incorporation in the PFC and hippocampus including the dentate gyrus. Aged rats (24 months old) showed a significant increase in aggrecan expression throughout the PFC and in the hippocampus particularly in the CA3 subfield, but not the dentate gyrus compared to young rats (5 months old), evaluated by the immunohistochemical method. YKS treatment decreased the age-related increase in aggrecan expression as well as normal expression in young rats. Aged rats also showed a decreased number of BrdU-labeled cells in the PFC and hippocampus, and these decreases were improved by YKS treatment, which also increased the numbers in young rats. These results suggest that aging influences the microenvironment for adult and immature neurons in the brain, which may affect the proliferation and migration of neural stem/progenitor cells, and YKS has pharmacological potency for these age-related events. These findings help to understand the physiology and pathology of the aged brain and provide an anti-aging strategy for the brain.** 

**Key words: aggrecan, perineuronal net, prefrontal cortex, hippocampus, bromodeoxyuridine, *Kampo* drug.**

Aging is considered as a complex multifactorial process that results in heterogeneous patterns of progressive morbidity and disability (Rowe and Kahn, 1987; Seeman and Robbins, 1994). In the brain, synaptic degeneration and neuronal loss as well as decreased neurogenesis are generally observed during the normal aging process in rodents, monkeys, and humans, but the mechanisms underlying these age-related deteriorations have not been fully understood.

The prefrontal cortex (PFC) is among the most sensitive to the influences of aging. Changes in PFC usually precede age-related changes in the majority of other cortical regions in individuals without dementia (Raz, 2000). These changes include a reduced volume of gray and white matter and decreased synaptic density (West, 1996; Raz, 2000; Tisserand et al., 2002). Functionally, normal aging consistently impairs PFC-dependent cognitions, including working memory, in humans (Nielsen-Bohlman and Knight, 1995; Schacter et al., 1996; Albert, 1997; Chao and Knight, 1997), monkeys (Rapp and Amaral, 1989; Herndon et al., 1997), and rats (Ando and Ohashi, 1991; Bimonte et al., 2003). Indeed, we have recently indicated that age-related working impairment is caused by reduced dopaminergic transmission in the PFC in rats (Mizoguchi et al., 2009). In addition to PFC cognitive deficits, dysfunction of the hippocampus, including the dentate gyrus, is considered an important phenomenon in the appearance of age-related memory impairment. Thus, the extent of memory impairment in aged animals is positively correlated with the magnitude of deterioration in several parameters reflecting the functional status of the hippocampus, such as glucose metabolism (Gage et al., 1984), electrophysiological properties (Barnes and McNaughton, 1985; Tombaugh et al., 2002), neurochemical markers (Gallagher et al., 1990), and cell morphology (Brizzee and Ordy, 1979; Geinisman et al., 1986). Moreover, the dentate gyrus has been demonstrated to produce new neurons during adulthood (Altman, 1962; Gross, 2000). This neurogenesis occurs in rodents as well as humans and declines with increasing age (Kuhn et al., 1996; Lemaire et al., 2000), in which the rate of neurogenesis in the dentate gyrus is positively correlated with memory performance of aged rats (Drapeau et al., 2003). Thus, the PFC and hippocam-

*Corresponding author. Tel: +81-562-46-2311; fax: +81-562-48-2373.
E-mail address: k-mizo@nils.go.jp (K. Mizoguchi).
*Abbreviations:* ANOVA, analysis of variance; BPSD, behavioral and psychological symptoms of dementia; BrdU, bromodeoxyuridine; CSPG, chondroitin sulfate proteoglycan; GAG, glycosaminoglycan; IgG, immunoglobulin G; MMP, matrix metalloproteinase; NSS, normal sheep serum; PBS, phosphate-buffered saline; PBS-T, PBS containing Triton X-100; PFC, prefrontal cortex; PNN, prereneuronal net; YKS, *yokukansan*.

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pus have received considerable attention in studies in the pathophysiology of the aging brain.

Chondroitin sulfate proteoglycans (CSPGs) are a member of a family of extracellular matrix-associated glycoproteins and are abundant in the CNS. In the developing and adult CNS, several reports have shown that CSPGs are generally thought to inhibit axonal growth, migration, and synaptic plasticity of neurons and to have a barrier function during regeneration (Perris and Johansson, 1990; Friedlander et al., 1994; Yamaguchi, 2000; Moon et al., 2001; Bradbury et al., 2002; Monnier et al., 2003; Rauch, 2004; Schwartz and Domowicz, 2004; Sivasankaran et al., 2004). Various CSPGs accumulate around neurons during postnatal development, forming so-called perineuronal nets (PNNs; Yamaguchi, 2000), which have a role in restricting experience-dependent plasticity in the adult brain, and the timing of PNN formation correlates with the ending of critical periods in the spinal cord and visual system (Guimarães et al., 1990; Kalb and Hockfield, 1990a,b; Pizzorusso et al., 2002). A recent study has revealed that aggrecan, a type of CSPG, is a crucial element in the initial assembly of PNNs (Carulli et al., 2007) and is a key molecule for modulating activity-dependent synaptic plasticity in the mouse barrel cortex (McRae et al., 2007). Aggrecan has also been shown to inhibit strongly neurite outgrowth enhanced by nerve growth factor in adult dorsal root ganglion neurons (Zhou et al., 2006) and is downregulated when neural stem/progenitor cells differentiate (Kabos et al., 2004). Thus, aggrecan creates an exceptional microenvironment in the brain, but the influence of aging on its expression has been unclear.

On the other hand, drugs obtained from natural plant sources are considered to be safe alternatives for the treatment of neuropsychiatric disorders (Kienzle-Horn, 2002; Carlini, 2003), and several traditional herbal medicines (called *Kampo* drugs in Japan, remedies composed of specified mixtures of crude drugs derived from plant, animal, and mineral materials) are effective in the field of neuropsychiatry (Kanba et al., 1998; Beaubrun and Gray, 2000). In particular, a *Kampo* drug *yokukansan* (YKS, *yi-gan san* in Chinese) has been widely used in a variety of clinical situations for treating several symptoms associated with age-related neurodegenerative disorders, such as behavioral and psychological symptoms of dementia (BPSD) (Iwasaki et al., 2005; Mizukami et al., 2009; Monji et al., 2009) and sleep disturbance in patients with dementia (Shinno et al., 2007), with very few side effects. Although scientific evidence of the clinical effects of this drug is limited, several basic researches have demonstrated that YKS improves learning disturbance and aggression in a rat model of Alzheimer's disease (Sekiguchi et al., 2009; Tabuchi et al., 2009) and ameliorates beta-amyloid-induced neurotoxicity *in vitro* (Tateno et al., 2008). These findings have suggested that YKS may be effective for age-related structural degeneration of cognition-responsible brain regions, such as the PFC and hippocampus.

Considering the literature, the present study was undertaken to investigate whether aging influences the microenvironment for functions of neural stem/progenitor cells in the brain and whether YKS has improving activity for age-related microenvironmental changes. For this purpose, we examined the expression of aggrecan, a major molecule of CSPGs, in the PFC and hippocampus of aged rats, and compared that to that of young rats. Next, the effects of YKS treatment on aggrecan expression in both rats were examined. Moreover, the influence of aging and the effects of YKS treatment on the number of neural stem/progenitor cells identified by bromodeoxyuridine (BrdU) incorporation were examined to estimate the functional significance of aging and regulating actions of YKS on aggrecan expression.

## EXPERIMENTAL PROCEDURES

### Animals

All animal experiments were performed in accordance with our institutional guidelines after obtaining permission from the Laboratory Animal Committee at the National Center for Geriatrics and Gerontology. The experiments in the present study were designed to minimize the number of animals used and to minimize their suffering.

Naive male F344/N rats were used. Two months-old rats weighing 200–250 g were purchased from Japan SLC Co. (Shizuoka, Japan) and used as young rats. Twenty-one months-old rats weighing 400–450 g were obtained from the Aging Farm, which was established at our institute in 2000 (Tanaka et al., 2000), and used as aged rats. The aged rats were produced by keeping for 18 months in the Aging Farm after purchase from the same breeder at 3 months old. They were housed two per cage in a temperature (23±1 °C) and light (12 h light/dark schedule; lights on at 8:00 AM)-controlled environment and were fed laboratory food and water *ad libitum*.

### Drug administration

YKS was supplied in the form of a water-extracted dried powder manufactured from a mixture of the crude drugs (Tsumura Co., Tokyo, Japan) (Table 1). The concentration of several effective chemicals was defined for each crude drug as an internal standard in the company's guide to Good Manufacturing Practices. The powdered drug was mixed at a concentration of 3% w/w with food pellets, made by Japan Clea Co. (Tokyo, Japan).

Young and aged rats were divided into two groups. One group was fed standard pellet chow for rodents (CE-2; Clea Japan) as the control group, and the other group was fed CE-2 containing YKS as the treatment group. The feeding period was 3 months; thus, at the end of the feeding, young rats were 5 months old and aged rats were 24 months old, respectively.

**Table 1.** Crude drug composition of *yokukansan* (YKS)

| Plant name | Composition (g) | Major components |
|---|---|---|
| *Atractylodis Lanceae Rhizoma* | 4.0 | Atractylodin |
| *Poria* | 4.0 | Eburicoic acid |
| *Cnidii Rhizoma* | 3.0 | Cnidilide |
| *Uncariae Uncis cum Ramulus* | 3.0 | Rhynchophylline |
| *Angelicae Radix* | 3.0 | Ligustilide |
| *Bupleuri Radix* | 2.0 | Saikosaponin a, c, d, e |
| *Glycyrrhizae Radix* | 1.5 | Glycyrrhizin |

### CSPG immunohistochemistry

Immunostaining for CSPG was performed according to previous reports (Carulli et al., 2007; McRae et al., 2007) with some modifications. Briefly, rats were perfused transcardially with 100 ml of 0.1 M phosphate-buffered and heparinized saline (pH 7.4), followed by 300 ml of 4% paraformaldehyde in 0.1 M phosphate buffer (pH 7.4) under pentobarbital anesthesia (50 mg/kg, i.p.). The brains were post-fixed for 1 h at 4 °C in the same fixative, cryoprotected in 30% sucrose before being frozen in powdered dry ice, and stored at −80 °C. Sections (30 $\mu$m) were cut using a freezing microtome. Free-floating tissue sections were rinsed three times with 50 mM phosphate-buffered saline (PBS, pH 7.4) containing 0.1% Triton X-100 (PBS-T). Endogenous peroxidase activity was blocked with 3% hydrogen peroxide in PBS-T, followed by incubation with 2× Block Ace solution (Dainippon Pharmaceutical, Osaka, Japan) for 30 min and with 10% normal sheep serum (NSS) in PBS-T for 30 min. The sections were then incubated overnight at room temperature with monoclonal mouse antibody against CSPG (Clone Cat-301, which recognizes CSPG1 (aggrecan); Millipore (Chemicon International), CA, USA), diluted 1:250 in PBS-T containing 1% NSS. After incubation, the sections were washed three times with PBS-T, and were incubated for 1 h at room temperature with peroxidase-linked sheep anti-mouse IgG (GE Healthcare UK, Buckinghamshire, UK), diluted 1:150 in PBS-T. Control sections, in which the mouse anti-CSPG antibody was replaced with NSS at the same dilution, were routinely processed together with the sections of interest. Peroxidase was visualized using 0.05% diaminobenzidine hydrochloride and 0.005% hydrogen peroxide.

The number of CSPG-positive cells was counted under a microscope, and the mean value of six counting data equaled the mean number of cells of each animal. Data are expressed as the number of CSPG-positive cells per area (the motor cortical region of the PFC or the CA3 subfield of the hippocampus).

### BrdU labeling and immunohistochemical detection

Rats were injected with a single dose of 5-bromo-2′-deoxyuridine (BrdU, 50 mg/kg i.p.; Sigma Aldrich, St. Louis, MO, USA) dissolved in sterilized saline at the end of the drug treatment period, according to a previous report (Montaron et al., 2006). The injection was repeated 24 h later.

Twenty-four hours after the second injection, the brains of rats were processed using the same procedure described above, except that the time of post-fixation was extended to 24 h. For immunohistochemical detection of BrdU-labeled cells, free-floating tissue sections (30 $\mu$m) were rinsed with PBS-T, treated with 2N HCl for 30 min at 37 °C, and then rinsed with PBS-T. After the procedures of endogenous peroxidase inhibition and blocking, the sections were then incubated overnight at room temperature with mouse monoclonal anti-BrdU antibody (Dako, CA, USA), diluted 1:200. The sections were incubated for 1 h with peroxidase-linked sheep anti-mouse IgG (GE Healthcare UK), diluted 1:150, and peroxidase was visualized. The sections were then stained with 2-fold diluted hematoxylin for nuclear staining.

The number of BrdU-labeled cells was counted under a microscope, and the mean value of six counting data equaled the mean number of cells of each animal. Data are expressed as the number of BrdU-labeled cells per area (the motor cortical region of the PFC, the CA3 subfield of the hippocampus, or the dentate gyrus).

### Statistics

All data were initially analyzed using one-way analysis of variance (ANOVA). Individual between-group comparisons were employed using Fisher's protected least significant difference test.

## RESULTS

### Influence of aging and YKS treatment on aggrecan expression

The results of immunohistochemical detection for aggrecan in the PFC are shown in Fig. 1. In young control rats (A), aggrecan-positive cells were distributed throughout the PFC, including the prelimbic, cingulate, motor, frontal, agranular insular, and orbital cortices. Fig. 1B shows the motor cortical region, and aggrecan immunoreactivity was present densely in dendrites and cell bodies of large cells that are thought to be neurons, in which aggrecan formed typical PNNs. Similar results were obtained from other cortical regions. Next, we examined the influences of aging on aggrecan expression, and found that the number of aggrecan-positive cells was markedly increased in aged control rats compared to young control rats (A and B vs. E and F). We further examined the effects of YKS treatment on the normal expression of aggrecan in young rats and on the age-related increased expression in the PFC. Aggrecan expression in young rats was reduced by YKS treatment compared to young control rats (A and B vs. C and D), and the aged-related increased expression was markedly attenuated by YKS treatment compared to aged control rats (E and F vs. G and H).

In the hippocampus (Fig. 2), aggrecan-positive cells were distributed in and around the pyramidal cell layer from CA1 to CA4 subfields, but were less common in the dentate gyrus (A). Fig. 2B shows the CA3 subfield, and aggrecan immunoreactivity was present densely in dendrites and cell bodies of pyramidal and non-pyramidal neurons, in which aggrecan also formed PNNs. Similar results were obtained from other subfields. The influences of aging and the effects of YKS treatment on aggrecan expression were similar to the results from the PFC. Thus, aggrecan expression was markedly increased in aged rats compared to young rats (A and B vs. E and F), and YKS treatments attenuated the expression in both young and aged rats (young, A and B vs. C and D; aged, E and F vs. G and H).

These immunohistochemical data were quantified, and the results are shown in Fig. 3. The number of aggrecan-positive cells in the motor cortical region of the PFC and hippocampal CA3 subfield was significantly increased in aged control rats compared to young control rats (PFC, $F(3,18)=36.215$, $P<0.001$; hippocampus, $F(3,18)=22.268$, $P<0.001$). YKS treatment significantly decreased the number of aggrecan-positive cells in both regions of young rats (PFC, $F(3,18)=36.215$, $P<0.05$; hippocampus, $F(3,18)=22.268$, $P<0.05$), and also strongly and significantly attenuated the age-related increased numbers in both regions (PFC, $F(3,18)=36.215$, $P<0.001$; hippocampus, $F(3,18)=22.268$, $P<0.01$). The influences of aging and the effects of drug treatment did not show region specificity in the PFC and hippocampus.

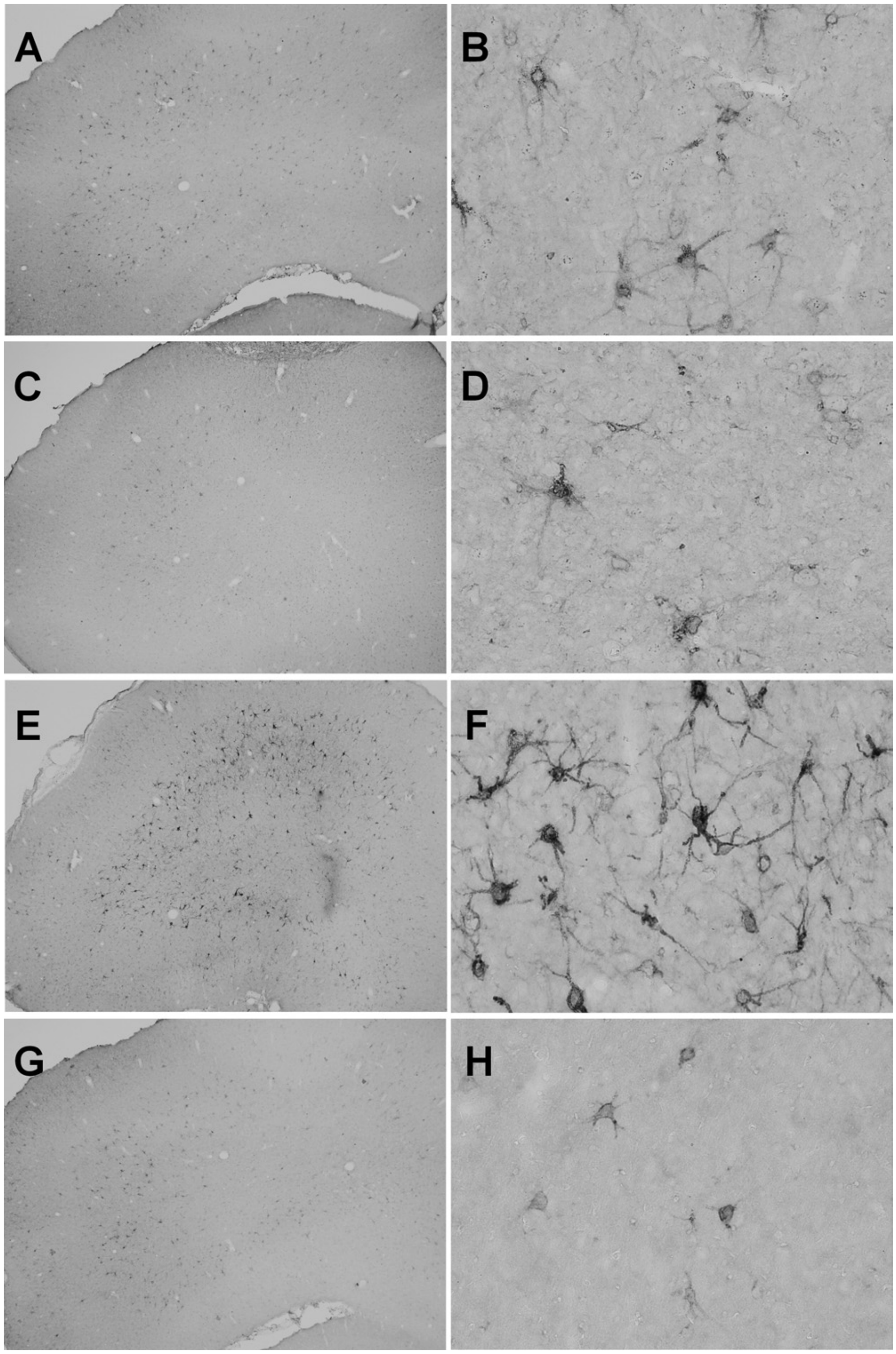


**Fig. 1.** Representative photomicrographs showing the influence of aging and the effect of YKS treatments on aggrecan expression in the rat PFC. (A, B) young control rats; (C, D) young rats treated with YKS; (E, F) aged control rats; (G, H) aged rats treated with YKS. The motor cortical region of the PFC is magnified (B, D, F, H). Original magnification: (A, C, E, G) ×40; (B, D, F, H) ×400.

### Influence of aging and effect of YKS treatment on the number of BrdU-labeled cells

The influences of aging and the effects of YKS treatments on the number of BrdU-labeled cells in the motor cortical region of the PFC, hippocampal CA3 subfield, and dentate gyrus were examined, and the results are shown in Figs. 4, 5. Fig. 4 shows representative photomicrographs indicating BrdU-labeled cells in the motor cortical region of the PFC (A, D, G, and J), hippocampal CA3 subfield (B, E, H, and K), and dentate gyrus (C, F, I, and L). BrdU-labeled cells

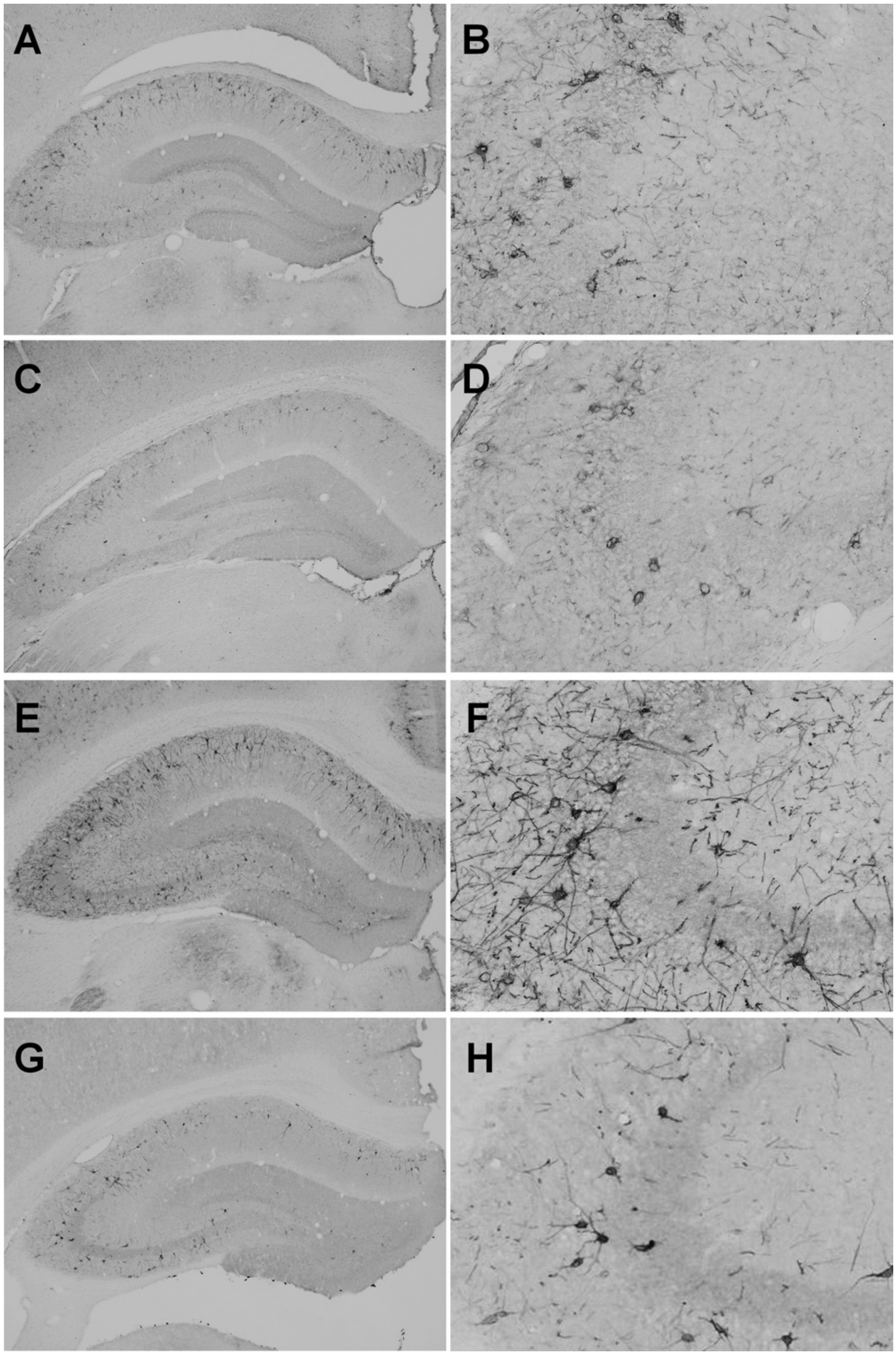


**Fig. 2.** Representative photomicrographs showing the influence of aging and the effect of YKS treatments on aggrecan expression in the rat hippocampus. (A, B) young control rats; (C, D) young rats treated with YKS; (E, F) aged control rats; (G, H) aged rats treated with YKS. The hippocampal CA3 subfield is magnified (B, D, F, H). Original magnification: (A, C, E, G) ×40; (B, D, F, H) ×400.

were small, and this morphology was in agreement with many reports indicating neural stem/progenitor cells in the brain (e.g. Montaron et al., 2006).

In the motor cortical region of the PFC, BrdU-labeled cells were decreased in aged control rats compared to young control rats (A vs. G), and the difference was significant ($F(3,12)=31.991$, $P<0.01$). Similar results were obtained in the hippocampal CA3 subfield and dentate gyrus; BrdU-labeled cells were significantly decreased in aged control rats compared to young control rats (CA3,

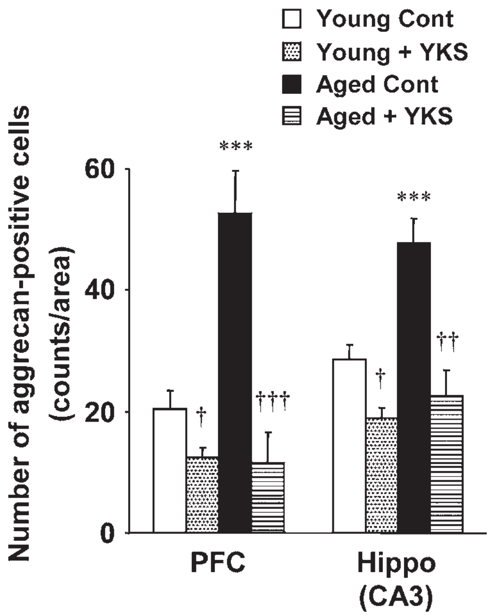


**Fig. 3.** Influence of aging and effect of YKS treatments on the number of aggrecan-positive cells in the motor cortical region of the PFC and in the CA3 subfield of the hippocampus of rats. Each column is the mean±SEM of six rats per group. Asterisks indicate a significant difference between young control and aged control rats, *** $P<0.001$. Daggers indicate a significant difference between control and YKS-treated in young or aged rats, $^{\dagger}$ $P<0.05$, $^{\dagger\dagger}$ $P<0.01$, $^{\dagger\dagger\dagger}$ $P<0.001$.

B vs. H, $F(3,12)=11.325$, $P<0.05$; dentate gyrus, C vs. I, $F(3,12)=21.716$, $P<0.05$). In the drug treatment study, BrdU-labeled cells were significantly increased by YKS treatment in all brain regions tested in young rats (PFC, A vs. D, $F(3,12)=31.991$, $P<0.001$; CA3, B vs. E, $F(3,12)=11.325$, $P<0.01$; dentate gyrus, C vs. F, $F(3,12)=21.716$, $P<0.001$). In particular, cells sometimes formed a cluster in the granular cell layer of the dentate gyrus (F). Aged rats showed similar results; BrdU-labeled cells were significantly increased by YKS treatment in all brain regions tested (PFC, G vs. J, $F(3,12)=31.991$, $P<0.01$; CA3, H vs. K, $F(3,12)=11.325$, $P<0.01$; dentate gyrus, I vs. L, $F(3,12)=21.716$, $P<0.05$). The influences of aging and the effects of the drug did not show region specificity in the PFC and hippocampus, including the dentate gyrus.

## DISCUSSION

Many molecular species of CSPGs are expressed in the CNS and have been shown to be involved in various cellular events in the formation and maintenance of the neural network (Bandtlow and Zimmermann, 2000; Oohira et al., 2000). In the present study, we observed that aging increased aggrecan expression and decreased the number of BrdU-labeled cells in the PFC and hippocampus, and that a *Kampo* drug YKS improved the age-related increase in aggrecan expression and decreased cell numbers; YKS also decreased normal aggrecan expression and increased cell numbers in young rats.

Our histological results (Figs. 1, 2) showed that aggrecan formed PNNs in the PFC and hippocampus, which is in line with previous studies showing that aggrecan was expressed in large neurons that were surrounded by PNNs during development and at adult in rodents (Carulli et al., 2006, 2007), and exhibited a PNN pattern in the mouse cerebral cortex (McRae et al., 2007).

Since normal aging is widely thought to be accompanied by changes in the structure and functions of the brain, we examined the influence of aging on aggrecan expression in the PFC and hippocampus. As shown in Figs. 1–3, in aged rats, aggrecan expression was markedly increased in the deep layer of the PFC and the CA3 subfield of the hippocampus, but not the dentate gyrus. Several reports have indicated that PNNs surround cell bodies and proximal dendrites of GABAergic interneurons, and consist of several kinds of hyaluronan, cell adhesion molecules, and CSPGs including aggrecan (Fox and Caterson, 2002), and that cells with PNNs mostly express the $GABA_A$ receptor $\alpha$1 subunit (Wegner et al., 2003). In addition, CSPG detected by the monoclonal antibody Cat-301 used in the present study has been characterized as an activity-dependent, neuronal surface proteoglycan (Zaremba et al., 1989). Therefore, increased expression of aggrecan might affect the neuronal activity of GABAergic neurons in the PFC and hippocampus. In the drug treatment study, YKS decreased not only the age-related increase in aggrecan expression in the PFC and hippocampus, but also the normal expression in young rats, suggesting that the decreasing activity of YKS is independent of age-related events. Regarding these results, two possible interpretations should be considered; one is that YKS decreases expression of aggrecan itself and the other is that YKS increases merely glycosylation levels in aggrecan structure, because intensity of Cat-301 immunoreactivity can be blocked by glycosaminoglycan (GAG) chains, sialic acid residues, and other O- and N-linked glycans, but the brain regions affected by changing glycosylation levels is restricted (Matthews et al., 2002).

In general, aged subjects of rats as well as monkeys and humans exhibit disturbances of PFC-and/or hippocampus-dependent brain functions, including cognition, learning, and memory, which are thought to be caused by reduced plasticity of neurons in these brain regions. In the cat visual system, PNNs detected with Cat-301 appear in postnatal development and are coincident with a reduction in synaptic plasticity, suggesting that plasticity of neurons after critical periods is regulated by changing aggrecan expression (Sur et al., 1988). In addition, regeneration of axons to recover from injury is limited by restricted plasticity (Kwok et al., 2008). Therefore, increased aggrecan expression in the aged rat brain (Figs. 1, 2) might be involved in age-related reductions in neuronal plasticity, and decreasing actions of YKS on the increased aggrecan expression might modify plasticity. In our recent study, the same aged rats used in the present study revealed working memory impairment caused by reduced dopaminergic transmission in the PFC (Mizoguchi et al., 2009), and indeed, YKS has potential to improve this working memory impairment (Mizoguchi, unpublished observations).

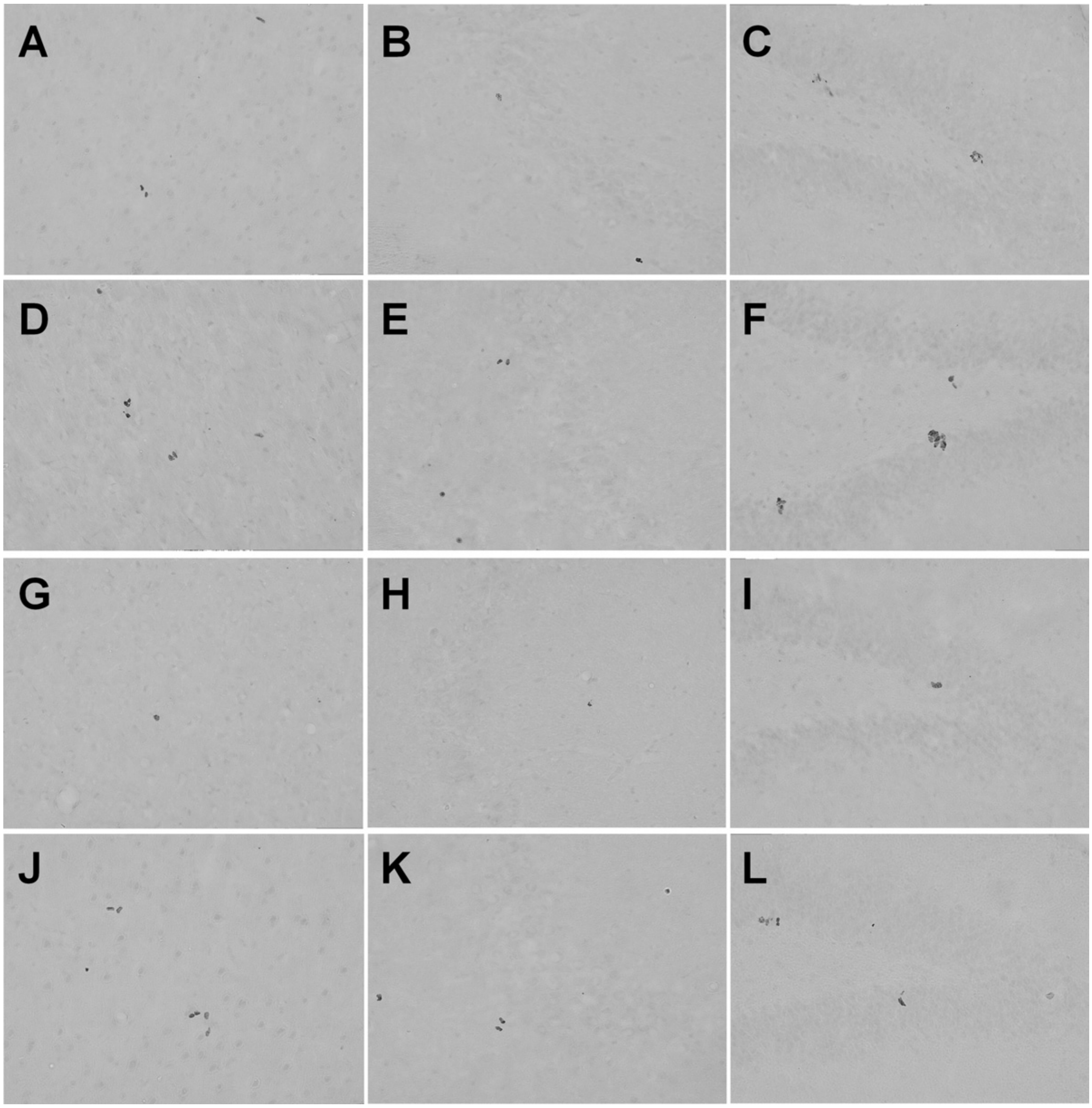


**Fig. 4.** Representative photomicrographs showing the influence of aging and the effect of YKS treatments on the number of BrdU-labeled cells in the PFC, hippocampus, and dentate gyrus of rats. (A, D, G, J) motor cortical region of the PFC; (B, E, H, K) hippocampal CA3 subfield; (C, F, I, L) dentate gyrus; (A, B, C) young control rats; (D, E, F) young rats treated with YKS; (G, H, I) aged control rats; (J, K, L) aged rats treated with YKS. Original magnification, ×200.

Although aggrecan in adult brains is mainly synthesized from neurons (Carulli et al., 2007), the mechanisms underlying increases in aggrecan expression in the aged brain are unknown. It is possible that degrading enzymes for aggrecan may be involved. For example, both matrix-degrading enzymes, matrix metalloproteinase 1 and 2 (MMP 1 and 2), can degrade aggrecan, while MMP2 also degrades versican (Fosang et al., 1992; d'Ortho et al., 1997; Passi et al., 1999), and their enzyme activity is involved in cell migration (Annabi et al., 2003; Son et al., 2006), but the influences of aging on these enzymes are unclear.

There are various CSPGs in the CNS, including aggrecan, neurocan, phosphacan, versican, and brevican, and these can act as barriers to cell migration (Grumet et al., 1996) and inhibit the migration of neural stem cells in the rat spinal cords *in vivo* (Ikegami et al., 2005). A previous study has also indicated that neural stem/progenitor cells participate in aggrecan as well as phosphacan in the construction of their own milieu by synthesizing these CSPGs and depositing them in their surroundings, suggesting that CSPGs may regulate their functions, such as proliferation, migration, and differentiation (Kabos et al., 2004). Considering that cell proliferation curtails with increasing age (Nacher et al., 2003) and the migration of new born cells also slows down (Heine et al., 2004), it was possible that the age-related increase in aggrecan expression affects the proliferation and migration of neural stem/progenitor

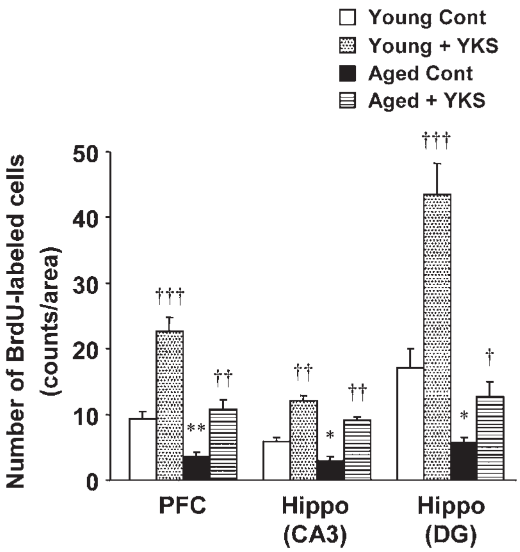


**Fig. 5.** Influence of aging and effect of YKS treatments on the number of BrdU-labeled cells in the motor cortical region of the PFC, the CA3 subfield of the hippocampus, and dentate gyrus of rats. Each column is the mean±SEM of six rats per group. Asterisks indicate a significant difference between young control and aged control rats, * $P<0.05$, ** $P<0.01$. Daggers indicate a significant difference between control and YKS-treated in young or aged rats, † $P<0.05$, †† $P<0.01$, ††† $P<0.001$.

cells in the PFC and hippocampus that includes the dentate gyrus, and that YKS treatment modifies these cell functions. Since age-related decline of the rate of neurogenesis is controlled at the level of proliferation and newly proliferated cells are efficiently detected by the BrdU incorporation method (Wu et al., 2008; Tang et al., 2009), we used this method to identify proliferating cells in the young and aged brains.

As shown in Figs. 4, 5, the number of BrdU-labeled cells in the motor cortical region of the PFC, CA3 subfield of the hippocampus, and dentate gyrus was decreased in aged rats. Although we did not further characterize these cells, BrdU-labeled cells observed in the dentate gyrus have well been characterized as newly proliferated cells expressing neural markers (Montaron et al., 2006; Wu et al., 2008) and our results regarding the dentate gyrus are in agreement with previous reports showing age-related declines of neurogenesis in the dentate gyrus and of hippocampus-dependent memory (Cameron and McKay, 1999; Montaron et al., 2006). Therefore, at least, BrdU-labeled cells in the dentate gyrus in our study are thought to be neural stem/progenitor cells. However, BrdU-labeled cells in the hippocampal CA3 subfield and PFC might not be neural stem/progenitor cells, because neurogenesis is thought to not occur in these regions. A reasonable interpretation is that these cells are glia such as astrocyte, microglia, and oligodendrocyte or other types of cells such as endothelial cells. On the other hand, referring that morphological features of these cells were very similar to those seen in the dentate gyrus and that stem cells can migrate generally to keep their proliferating activity, it is possible that BrdU-labeled cells detected in the hippocampal CA3 subfield and PFC are also proliferated cells migrated from the neurogenesis sites. In the drug treatment study, YKS produced significant increases in the number of BrdU-labeled cells in the young and aged brains, suggesting that the increase in the number of BrdU-labeled cells in the dentate gyrus is caused by increased proliferation of neural stem/progenitor cells and that in the CA3 and PFC may be due in part to increased migration from generating regions. These effects of YKS on cell functions are thought to occur in an age-independent manner. A generally accepted theory is that CSPGs are abundant in regions where neural stem/progenitor cells are absent, and that their segmental distribution inversely correlates with the migration pathway of cells during development (Perris et al., 1991). In the present study, the age-related increase or YKS-induced decrease in aggrecan expression was observed in migration regions (i.e. CA3 subfield and PFC), but not in proliferation region (i.e. dentate gyrus). Moreover, the changes in aggrecan expression and the number of BrdU-labeled cells were negatively correlated between aging influences and YKS effects. Considering that CSPGs including aggrecan inhibit cell migration (Perris and Johansson, 1990; Yamaguchi, 2000; Rauch, 2004; Schwartz and Domowicz, 2004), these findings lead the possibility that age-related or YKS-induced changes in aggrecan expression in the CA3 subfield and PFC (Figs. 1–3) are related to decreased or increased migration of cells into these regions. However, the age-related decrease or YKS-induced increase in cell proliferation in the dentate gyrus is not thought to be related to changed aggrecan expression there, because age and YKS did not affect its expression clearly.

Although age-dependent decline of neurogenesis is thought to be attributed to decreases in proliferation and growth factor signaling (Tropepe et al., 1997; Enwere et al., 2004; Shetty et al., 2005) and senescence of neural stem/progenitor cells (Molofsky et al., 2006), a recent study has indicated that neural stem/progenitor cells in the aged mouse brain retain their potential for proliferation, migration, and differentiation into neurons despite the number is lower (Ahlenius et al., 2009), suggesting that age-related changes in the microenvironment suppress the proper functions of neural stem/progenitor cells. Considering the neurite outgrowth inhibitory actions of CSPGs including aggrecan, the age-related increase in aggrecan expression might be a candidate for the changed microenvironment in the aged brain, which also might cause lower efficiency in cell attachment and development when neural stem cells are therapeutically grafted to improve disrupted function of the aged brain. Meanwhile, age-related decline of neurogenesis can be improved to adult levels by several approaches, such as enriched environment (Kempermann et al., 2002), infusion of growth factors (Jin et al., 2003), or antidepressant treatment (Drew and Hen, 2007). The present finding that a *Kampo* drug increased the proliferation of neural stem/progenitor cells in the dentate gyrus (Figs. 4, 5) provides additional information to enhance neurogenesis in young and aged brains.

Alternatively, bacterial enzyme chondroitinase ABC, which is a deglycosylation enzyme for GAG chains of CSPGs, is an useful tool to promote neurite outgrowth to repair CNS injury, because removal of CSPGs by its treatment enhances regeneration and plasticity of neuronal connections and improves motor and sensory functions in animal models of CNS injury (Krekoski et al., 2001; Bradbury et al., 2002; Caggiano et al., 2005; Fouad et al., 2005; Groves et al., 2005). However, since there are several problems with the use of this enzyme as a treatment option for CNS injury in humans (Tester et al., 2007), alternate approaches have been tired; for example, inhibition of GAG chain polymerization using short interfering RNA against chondroitin polymerizing factor that is the rate-limiting enzyme for CSPG biosynthesis (Laabs et al., 2007) or local production of this enzyme by adenoviral mediated gene transfer (Curinga et al., 2007). The decreasing activity of YKS on CSPG expression described in the present study will be applied to treat CNS deficits, such as neurodegeneration, stroke, and spinal cord injury in animals as well as humans.

## CONCLUSION

In conclusion, this paper shows that aging increased aggrecan expression in the PFC and hippocampus and a *Kampo* drug YKS decreased this increased expression in addition to normal expression in young rats. Moreover, YKS enhanced neurogenesis in young and aged brain. These findings provide important information for understanding the aging physiology and pathology of the brain and for an anti-aging strategy using *Kampo* drugs. We have great interest in the active compound(s) of YKS causing the pharmacological activity found in the present study.

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